

Cross-talker generalization: analysis and figure update

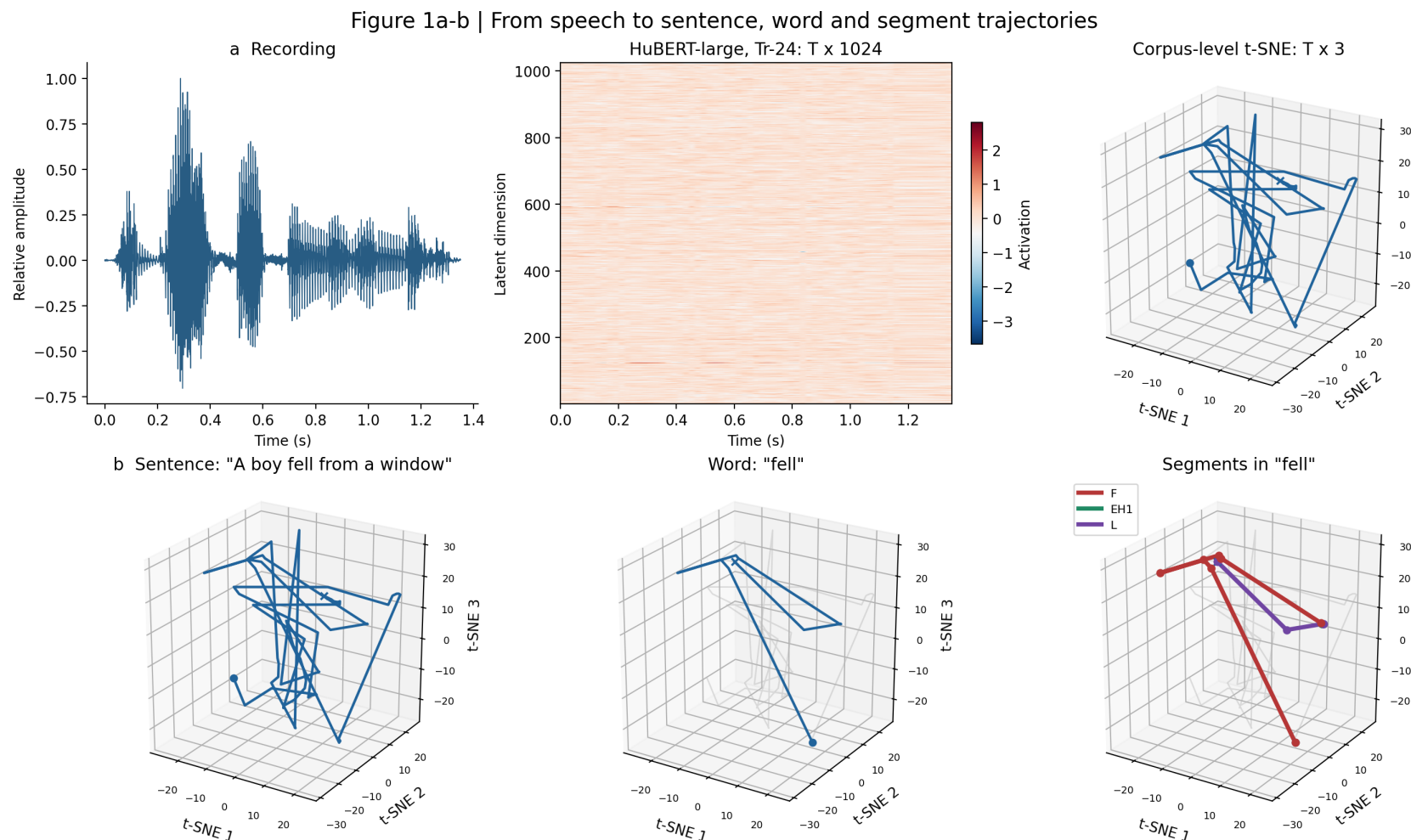
Prepared for Florian Jaeger by Zhengyang Jin | 6 September 2026 | For discussion

This report follows up on our September 1 meeting and your subsequent email. It separates newly calculated validation panels from previously fitted SBI/HVE results. It is a review package, not a claim that the manuscript is complete.

1. **Optimization and reporting.** Predictor-only likelihood and predictor z remain separate quantities. The principal SBI display is the layerwise z profile with three-fold intervals and a behavioral reference. Likelihood evaluation and the two condition comparisons remain complementary analyses; Tr-24 remains the common-layer summary discussed in the meeting.
2. **Consistent figures.** The available layerwise z figures, conditional S-curves and nested comparisons are linked with their fit scope. The revised HVE training-fold z still requires a compatible ceiling; the notebook test-refit ceiling cannot simply be reused.
3. **Figures 1 and 2.** This update supplies new Tr-24 representation/trajectory panels, checked language labels, a proposed PHOIBLE inventory panel, the requested 36-L2-talker matrix, and newly computed AN19/X21 control word-level similarity curves. AN19 has no existing phone-level labels, as Zhengyang confirmed; segment panels 2b/2d are deferred pending a separately agreed annotation effort.
4. **AN19 acoustic controls.** The first component analysis used different coordinate scaling from the full baseline. I have not interpreted its component ranking as a controlled ablation; that comparison needs a matched-preprocessing rerun.
5. **Exposure understanding.** The proposed extension should connect recognition of particular exposure sounds/words to related test words. A correlation between overall exposure and test accuracy would not answer that question.

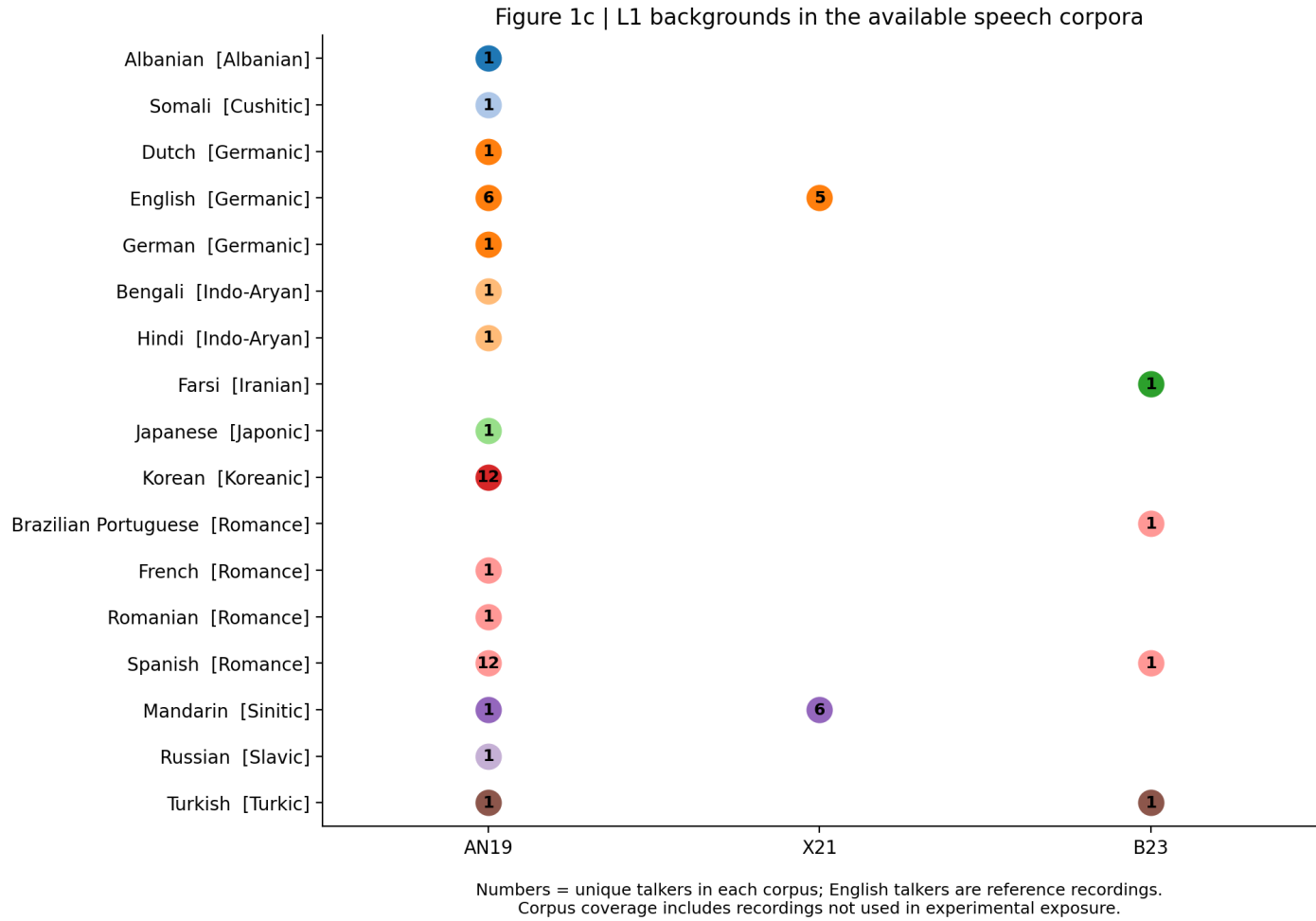
The following pages present the available panels and the small number of decisions/data transfers needed to finish the remaining work. All new figures have PNG/SVG exports and source tables.

Figure 1a-b: a verified HuBERT example at Tr-24



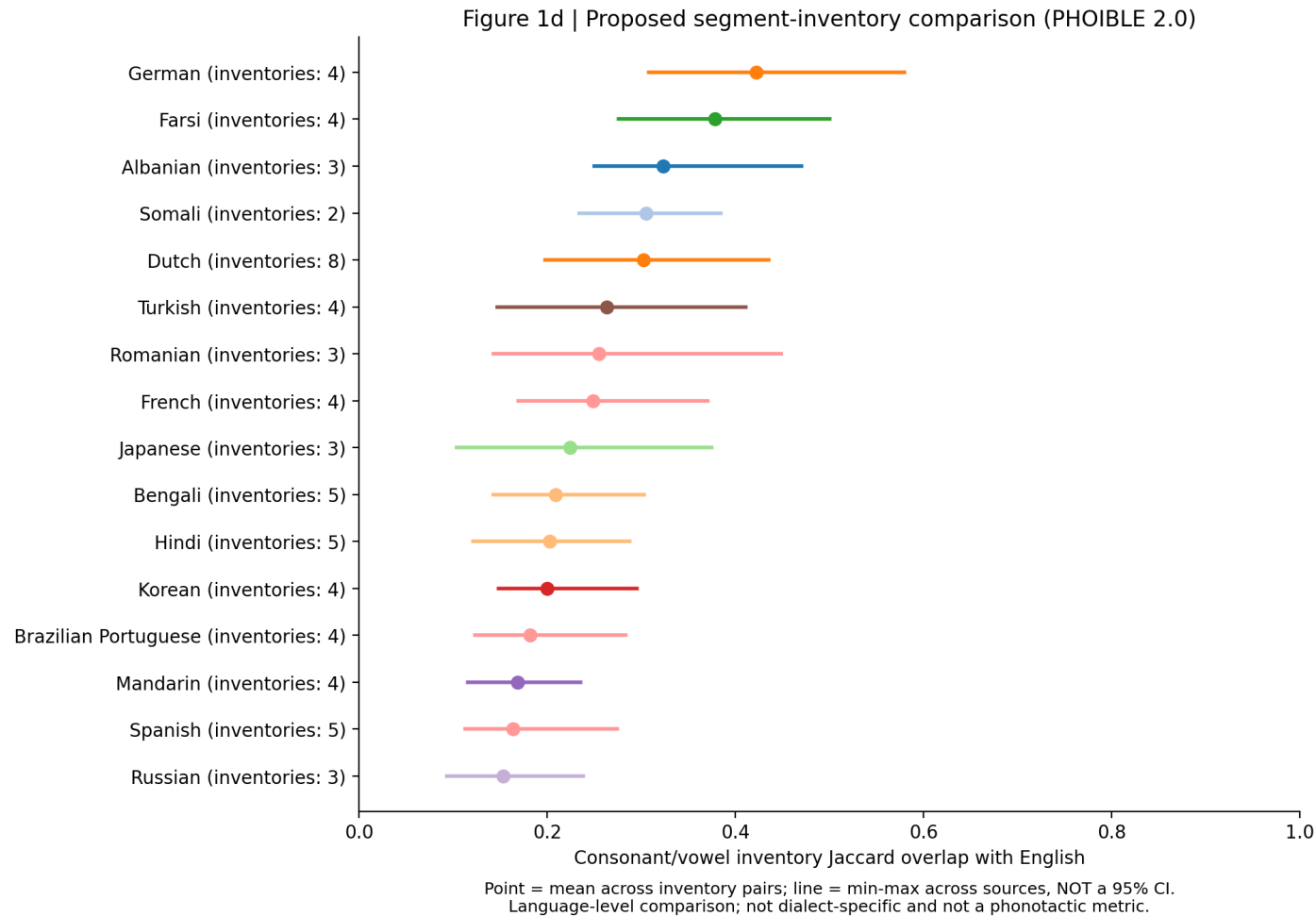
The recording is “A boy fell from a window,” produced by English-L1 male talker 055 in X21. Full and reduced features come from the same HuBERT-large checkpoint and layer: 67 frames, 1,024 full dimensions, and three t-SNE dimensions. The t-SNE fit covered all 660 corpus sentence segments, with seed 42; it was not refitted for this illustration. Word and phone intervals locate “fell” and F, EH1, L. A one-frame segment is shown as a point. All trajectories share coordinates and axis limits; no per-token normalization or synthetic smoothing is applied. Circles mark starts and crosses mark ends.

Figure 1c: language coverage with checked source labels



The labels “Indian” and “Chinese” in the AN19 manifest are resolved to Hindi and Mandarin using Table I of the original study; original labels remain in the source table. Colors indicate broad genealogical branches. Counts describe the available speech corpora, including English reference recordings, not unique participants or exposure conditions. This display contains 17 distinct L1 labels after harmonization across studies.

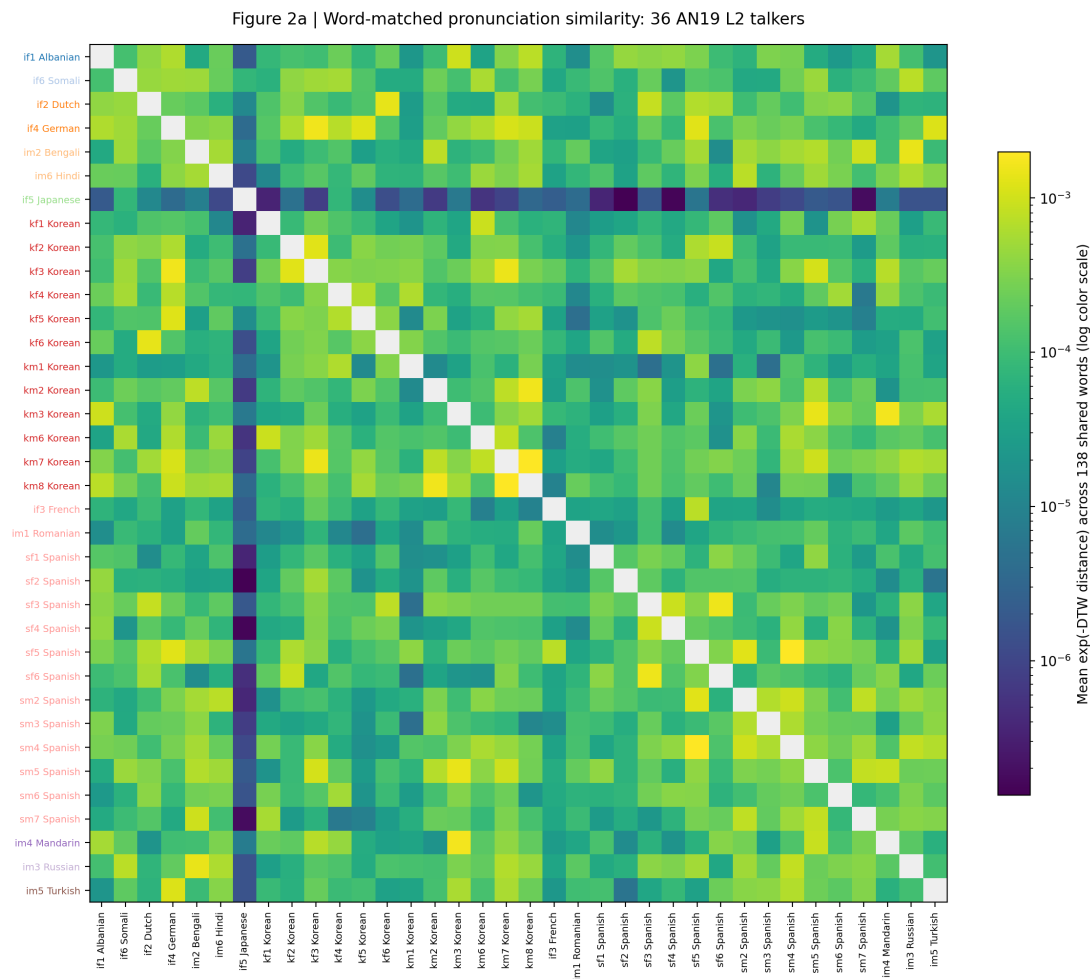
Figure 1d: a computed proposal for inventory similarity



Proposed metric, not yet the full captioned construct. For each language and English inventory pair, I computed Jaccard overlap of consonant/vowel symbols. Points average all available inventory pairs; lines show their min-max spread, not 95% confidence intervals. This is sensitive to inventory selection and transcription conventions, and is not a phonotactic measure or a speaker-specific dialect comparison. A decision is needed on whether this narrower panel is suitable or whether separate suprasegmental/phonotactic sources should be added.

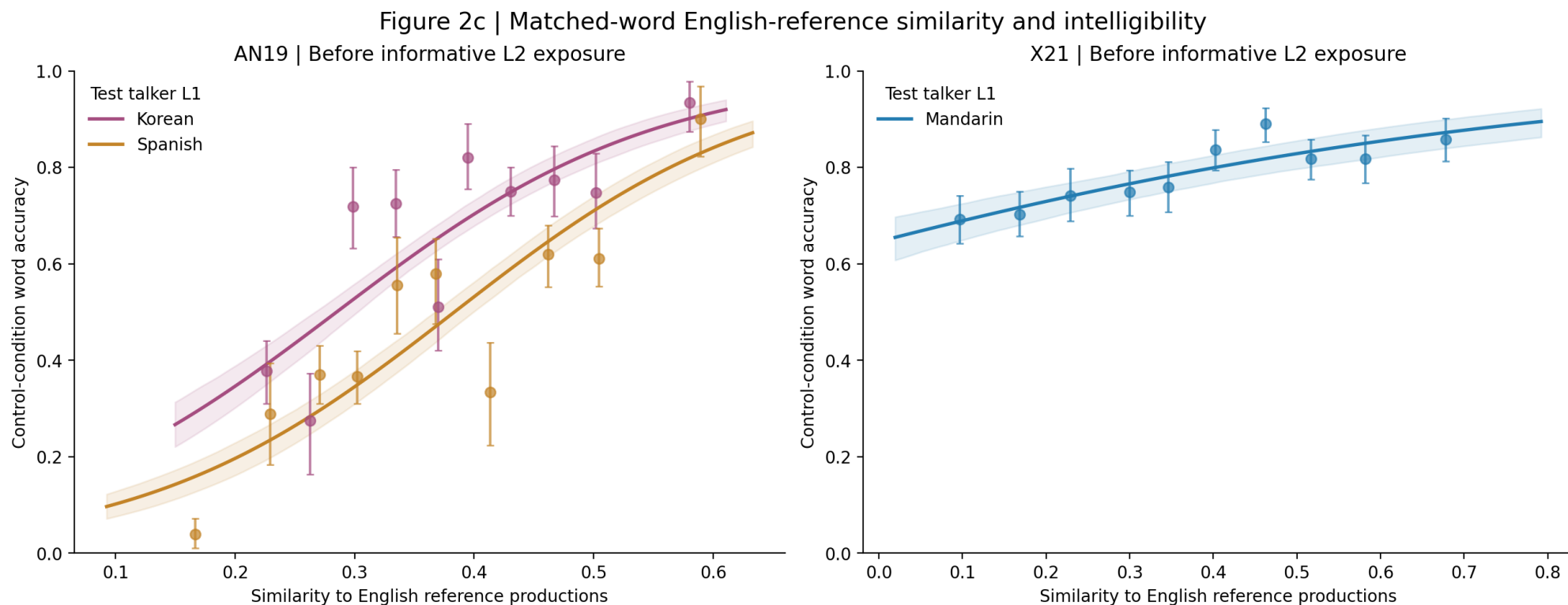
Source: Moran & McCloy (eds.), PHOIBLE 2.0, 2019, CC BY-SA 3.0. Data are from the fixed v2.0 release; inventory IDs and all comparisons are exported.

Figure 2a: pronunciation similarity among 36 AN19 L2 talkers



The original available matrix included 42 talkers: 36 L2 talkers and six English reference talkers. This panel selects the 36 L2 talkers requested in the caption and orders them by L1/branch. Each cell uses the same 138 shared words and the stored Tr-24 base-model calculation: equal-weight mean word similarity, $\exp(-\text{DTW distance})$, with mean-sequence-length DTW normalization and $\tau = 2$. The color scale is logarithmic; self-comparisons are masked. The six English talkers remain reference sources for Figure 2c rather than extra rows in this panel.

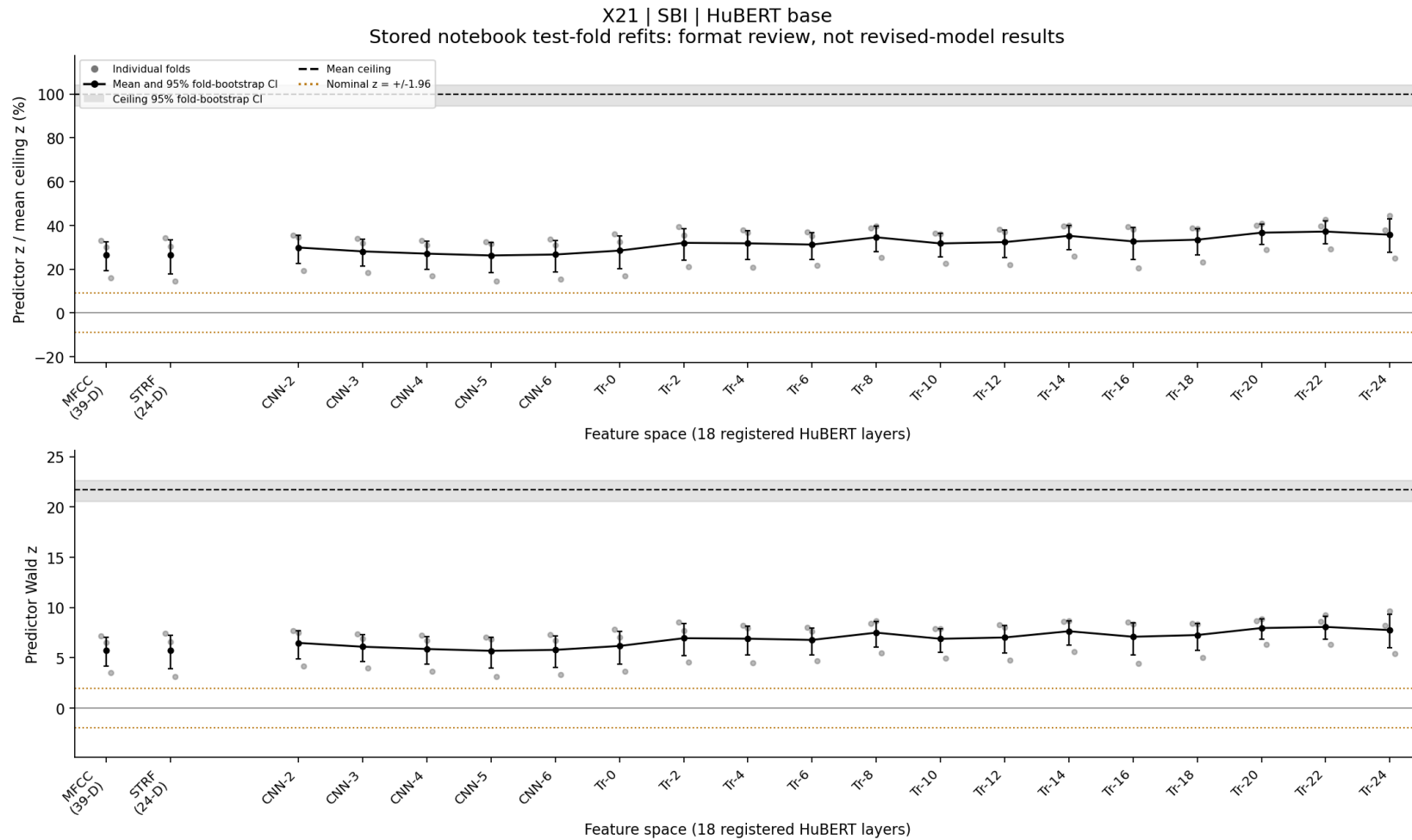
Figure 2c: English-reference similarity and control intelligibility



Each test-word trajectory is compared with the same word from six English talkers in AN19 or five in X21. X21 word intervals are cropped from the corresponding sentence features. Distances are averaged equally over English talkers. For a readable response-independent scale, similarity is $\exp(-d/s)$, where s is the dataset median distance across unique matched targets; this scale was not selected using accuracy.

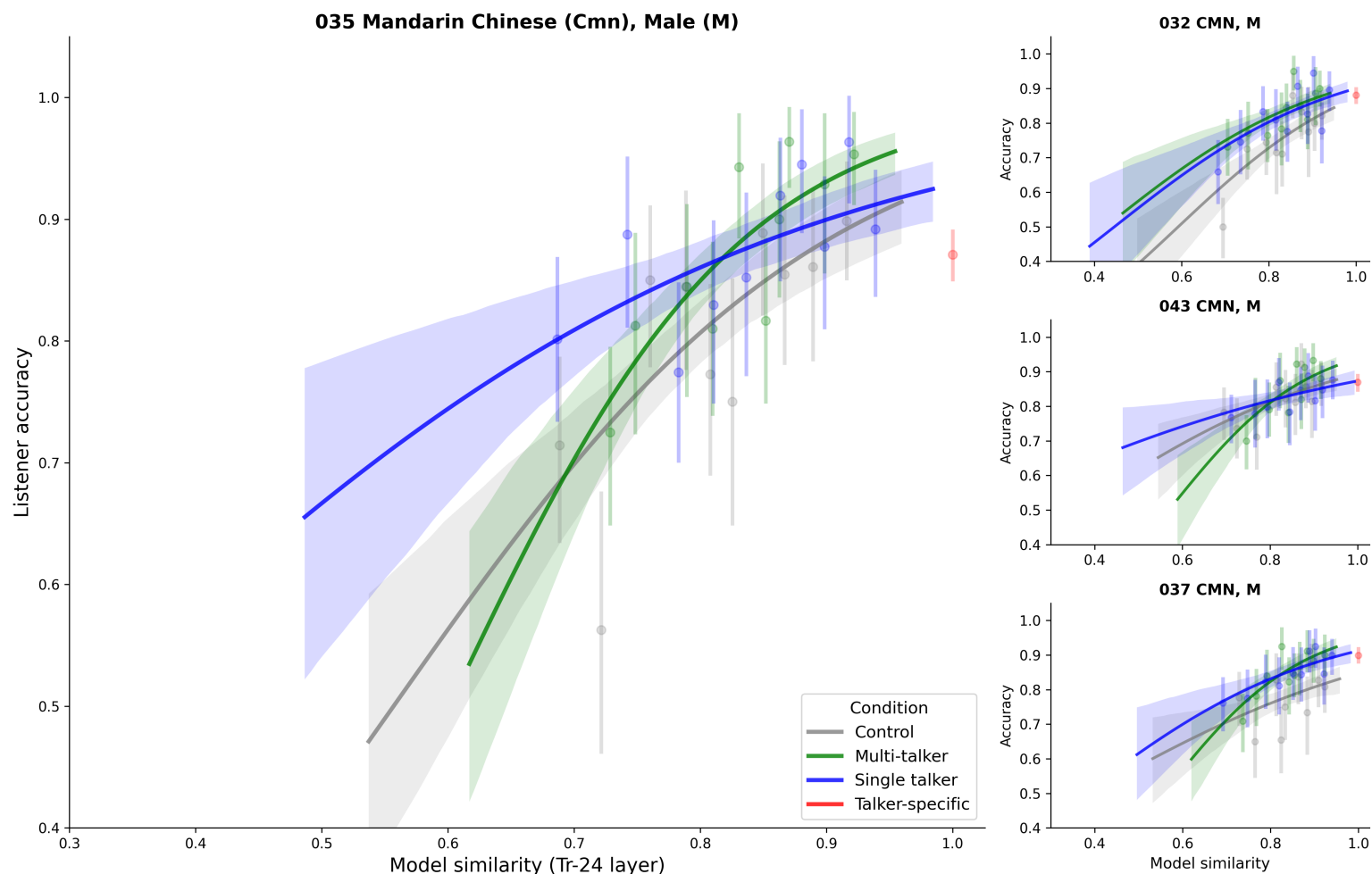
Points are ten within-accent quantile bins. Curves are descriptive binomial logistic fits; both use 95% participant-bootstrap intervals (1,000 resamples). The association is positive in all three plotted accent groups. These are not cross-validated GLMM tests; intervals condition on the sampled items/talkers. AN19 includes 1,880/1,920 control answers from 40 participants. The 40 "wave" answers are provisionally excluded: corresponding English hw74 files and features exist, but are labeled "wade", whereas all L2 files say "wave". This lexical-label conflict needs audio/content verification, not an assumption of missing recordings. X21 includes all 4,117 control answers from 80 participants. This panel does not estimate exposure-condition effects.

SBI layerwise z: the requested display, with its source identified



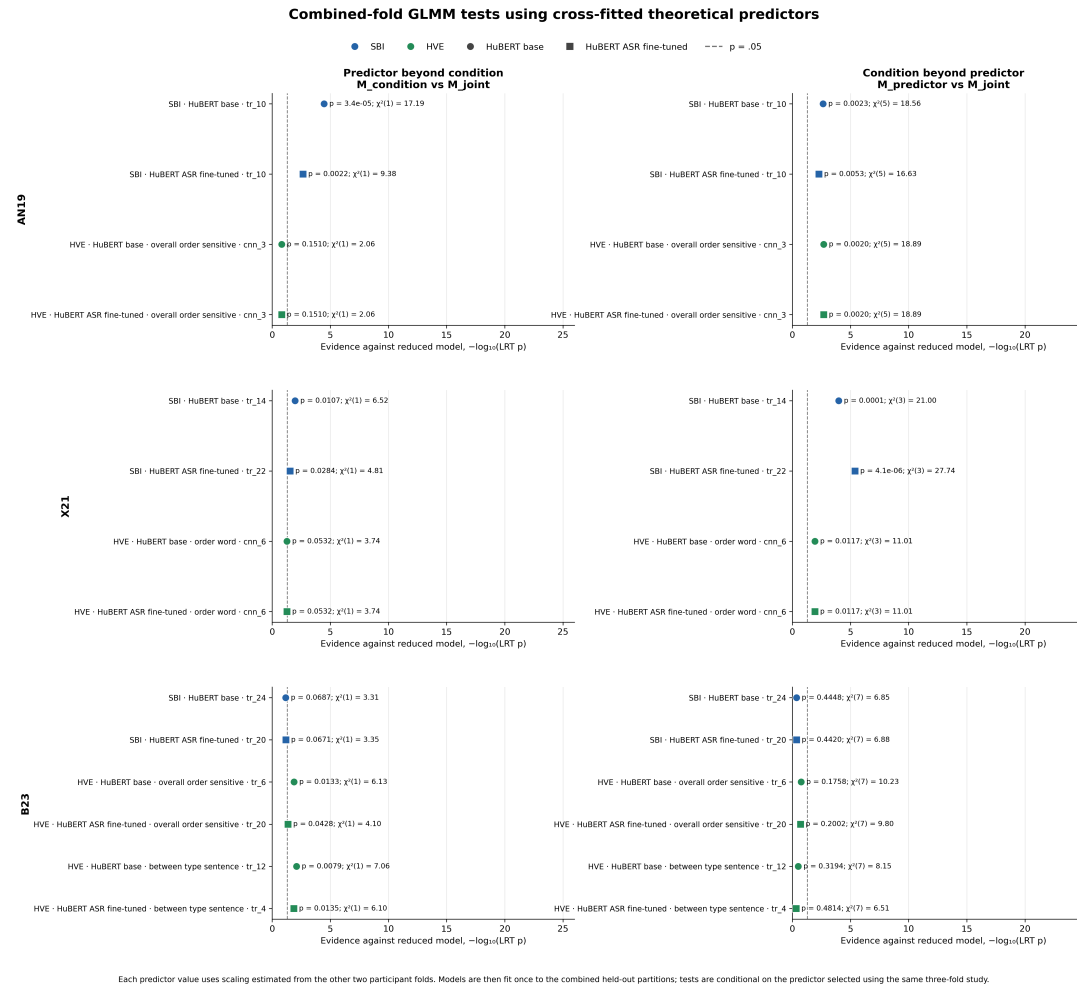
This X21 example illustrates the principal SBI display: acoustic baselines first, then registered layers; three fold points; mean and 95% fold-bootstrap intervals; a gray ceiling interval; and labeled nominal z references. The lower panel retains raw z. These are stored notebook test-fold-refit statistics, not newly fitted current-definition results. A z/ceiling ratio is not percentage variance explained. All datasets and variants, plus the separate revised HVE profiles, are available in the z gallery. The manuscript draft requests a Bonferroni reference; its testing family still needs to be specified.

Conditional curves: retain the X21 comparison discussed in the meeting



This existing descriptive panel preserves the multi-condition, multi-talker comparison in the recovered `Figure_fit_reg.ipynb`. It is distinct from the new control-only Figure 2c: its x-axis is exposure-test model similarity, not a fixed English-reference predictor for every condition. Curve offsets help visualize what may remain associated with condition; they do not themselves test whether condition adds to the theoretical predictor. The companion pooled-bin plot and all source tables remain available in the August 21 package.

Both condition comparisons remain necessary



The left comparison asks whether adding SBI/HVE to condition improves fit; the right asks whether adding condition to SBI/HVE improves fit. The theoretical predictor is not separately reoptimized for those two tests. These existing combined-fold likelihood-ratio results are supplementary association tests: predictor scaling is cross-fitted, but layer/method selection used the same study, so p-values are selection-conditional. They should not be described as independent confirmation or confused with frozen-model held-out scores. The displayed selected-layer tests are not the fixed-Tr-24 summary.

Remaining inputs and decisions

Item	What is already available	What is still needed
Figure 2b	AN19 word features, target transcripts, English references	Verified AN19 phone boundaries and an outcome-independent phoneme-selection rule
Figure 2d	Full caption and orthographic listener responses	Figure 2b deviations; pronunciation lexicon; response-to-phone alignment and minimal-pair-neighbor rule
Figure 1d	Computed PHOIBLE inventory proposal	Accept the narrower construct or specify additional sound-structure sources
Current z/ceiling	Revised HVE fold coefficients and older notebook z tables	Agree fit scope and rerun compatible current-definition ceilings; restore all-layer current SBI coefficients
Objective comparison	Likelihood/z selection implementation	Separate B23's 97 ordered and 168 order-independent participants, or use identical rows
Acoustic attribution	Full baselines and first component outputs	Recompute components with the full baseline's coordinate scaling and matched rows

Zhengyang confirmed that AN19 has no existing phone-level labels. Figures 2b/2d therefore cannot be completed from the current annotations and are deferred, not treated as files that merely need locating. Completing them would require a separately agreed phone-alignment and verification effort, plus response-error/lexical-neighbor definitions for 2d. The draft's generic statement about checked OSF annotations is not evidence of AN19 labels. The recovered notebooks supply S-curves and method illustrations only. Original audio can be available in the local Git LFS object store even when the working-tree WAV is a pointer; this does not supply phone boundaries.

The local manuscript additionally lists ASR fine-tuning, architecture, full-dimensional/reduced representations, UMAP, PCA and similarity-measure ablations. Those should be reconciled with the current Overleaf list before claiming complete figure coverage. The detailed implementation/source notes are separate from this report in SOURCE_REVIEW.md and the project TODO.